Table 1. Overall performance comparison of various approaches on LibriMix. Sys. {1-3} are the experimental results from ESPnet[1], Sys. {4-5} are the results of AED-based models, and Sys. {6-8} are the results of the LLM-based models.

Sys.	type	Speech Encoder	WER (%) ↓ dev	WER (%) ↓ test
1	ESPnet[1] Baseline	Whisper small	26.0	25.0
2		Conformer	24.7	23.3
3		+ WavLM Large upstream	19.4	17.1
4	AED	WavLM Base+	18.9	17.7
5		WavLM Large	10.6	9.2
6	LLM	WavLM Base+	17.6	15.9
7		WavLM Large	11.4	10.2
8		+ LibriMix Fine-tuning	**10.3**	**9.0**

Table 2. Performance comparison with and without LoRA fine-tuning in the case of different speech encoders.

Sys.	Speech Encoder	LoRA	WER (%) ↓ dev	WER (%) ↓ test
-	WavLM Base+	✗	19.4	17.3
Tab. 1, Sys. 6		✓	17.6	15.9
-	WavLM Large	✗	12.6	11.3
Tab. 1, Sys. 7		✓	11.4	10.2
-	+ LibriMix Fine-tuning	✗	10.8	9.5
Tab. 1, Sys. 8		✓	10.3	9.0

frozen. Sys. {6-8} in Table 1 are the results of the LLM-based approach proposed in this work. When using WavLM Base+ as the speech encoder, the LLM-based method (Sys. 6, Tab. 1) outperforms the AED-based method (Sys. 4, Tab. 1). However, when WavLM Large is used as the encoder, the AED-based method shows a significant performance boost (Sys. 5, Tab. 1), even surpassing the LLM-based method (Sys. 7, Tab. 1), which indicates that AED-based systems are more dependent on encoder performance. Initializing the LLM-based system with the speech encoder fine-tuned on LibriMix using AED method results in the best performance (Sys. 8, Tab. 1). Therefore, in the performance on LibriMix test set, the advantage of the LLM-based system over the AED-based system is not very pronounced (9.0% WER in Sys. 8 *vs.* 9.2% WER in Sys. 5). This is similar to conclusions drawn from single-speaker ASR studies [18, 21], as LibriMix is simulated data and contains only two speakers per utterance, making it less challenging compared to real conversational scenarios.

Table 2 shows the performance comparison of different speech encoders with and without LoRA fine-tuning. Similar to the conclusions in [18] and [22], introducing LoRA fine-tuning into the LLM consistently improves performance regardless of the speech encoder used. This indicates that LoRA fine-tuning can adapt the LLM output to the style of

Table 3. Performance comparison of freezing and jointly training the speech encoder with and without fine-tuning on LibriMix using AED method.

Sys.	LibriMix Fine-tuning	Freeze Encoder	WER (%) ↓ dev	WER (%) ↓ test
Tab. 1, Sys. 7	✗	✗	11.4	10.2
-		✓	47.8	46.7
-	✓	✗	11.4	10.1
Tab. 1, Sys. 8		✓	10.3	9.0

Table 4. Performance comparison of single-stage training and multi-stage training strategy. Multi-stage training refers to sequentially unfreezing and jointly training in the order of projector → speech encoder → LoRA. When the "Freeze Encoder" option in the table is set to True, the second stage is skipped. Single-stage training refers to jointly training all these modules from the beginning.

Sys.	Freeze Encoder	Training Strategy	WER (%) ↓ dev	WER (%) ↓ test
-	✗	single-stage	11.7	10.4
-		multi-stage	11.4	10.1
-	✓	single-stage	10.5	9.2
Tab. 1, Sys. 8		multi-stage	10.3	9.0

SOT-based multi-talker transcription. In [21], promising performance can be achieved even without introducing the LoRA adaptor, possibly because the transcription style of the single-talker Librispeech used in [21] is similar to the output of the original LLM.

Table 3 presents the impact of freezing the speech encoder during training. When the initialized speech encoder is not fine-tuned on LibriMix using the AED method, freezing the encoder results in poor performance because the encoder has not adapted to the LibriMix dataset. However, when using the encoder fine-tuned with LibriMix (Sys. 5, Tab. 1), freezing the encoder during training results in better performance. This is likely because the fine-tuned encoder already has excellent representation extraction capabilities on LibriMix and does not require further adjustment.

Fig. 3 shows the comparison of training curves in the first training stage, where only the projector module is trained, using either a fine-tuned encoder or a non-fine-tuned encoder. When using the fine-tuned encoder, the model quickly converges to a very high accuracy. In contrast, using the original WavLM model results in slower and less complete convergence. This indicates that if we have a high-quality encoder, simply aligning the modality of speech representations with the LLM can directly achieve a relatively good performance. Conversely, for an unadapted encoder, merely training the projector to perform alignment is insufficient, which is similar to the conclusion in Table 3.

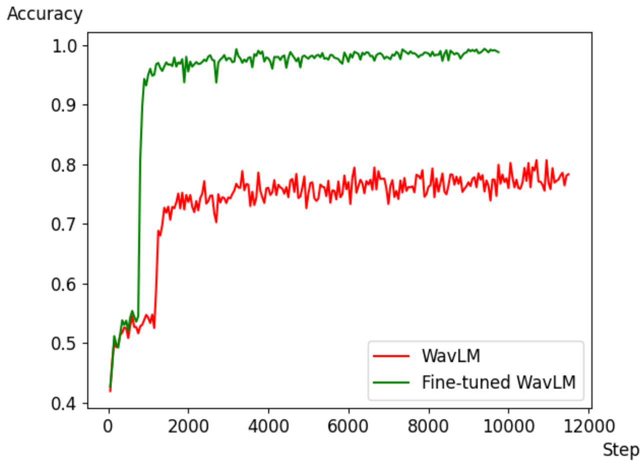

Fig. 3. Training accuracy of the next token prediction with the training steps in the first training stage, where only the Projector is involved in training. Different colored curves represent whether the speech encoder has been fine-tuned by LibriMix.

Table 4 presents the comparison between single-stage and multi-stage training strategies. The results show that, regardless of whether the speech encoder is frozen, multi-stage training outperforms single-stage training. This indicates that multi-stage training helps the model better align auditory and textual information.

3.2. Experiment with AMI

3.2.1. Experimental settings

To evaluate the LLM-based multi-talker ASR approach in a more realistic setting, we conducted experiments on real-world corpus AMI. The AMI meeting corpus includes approximately 95 hours of real-world meeting recordings, with the training, validation, and evaluation sets comprising 76.9, 8.9, and 8.7 hours, respectively. Each meeting involves 3 to 5 participants. The audio in the AMI corpus was recorded using an 8-channel microphone array, known as multiple distant microphones (MDM). Typically, the first channel is used for monaural ASR evaluation, referred to as the single distant microphone (SDM) setting. Additionally, the AMI corpus includes near-field single-speaker audio recorded by independent headset microphones (IHM) worn by each participant.

In this work, we conducted experiments using the SDM setting. However, in the original SDM, the audio is segmented by oracle timestamps into utterances containing only a single speaker. To evaluate SOT-based multi-talker ASR, we followed the approach in [11] to use *utterance group*-based evaluation. An *utterance group* is defined as a set of utterances connected by speaker overlap regions. Correspondingly, SOT-style transcriptions are generated in the order of the emission time of each speaker.

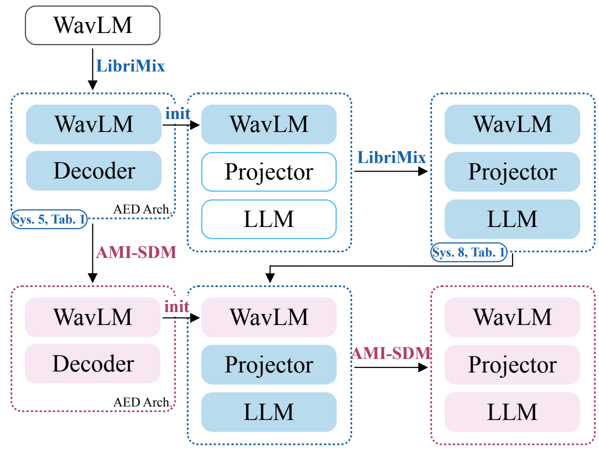

Fig. 4. An illustration of the training process of the proposed LLM-based multi-talker ASR system on the LibriMix (blue background) and AMI-SDM (pink background).

In addition to using simple WER for evaluation, we also introduced the concatenated minimum-permutation word error rate (cpWER) [35] for comparison with previous work [11, 36]. In each *utterance group*, as shown in Fig. 1, the transcriptions of the same speaker are concatenated, and the minimum WER across all possible speaker permutations is taken as the cpWER.

For the training details, as shown in Fig 4, we first fine-tuned the WavLM AED model, which was pre-trained on LibriMix (Sys. 5, Tab. 1), using the AMI-SDM *utterance group* segments. Subsequently, we integrated this fine-tuned WavLM encoder into the best-performing system from the LibriMix experiment (Sys. 8, Tab. 1) and further fine-tuned it on the AMI-SDM *utterance group* segments. The training strategy and configuration remained consistent with those employed in the LibriMix experiment.

3.2.2. Experimental results

The overall experimental results on the AMI-SDM evaluation set are presented in Table 5. Sys. {1-3} are from previous work, all relying on large-scale supervised data for pre-training. As shown by the experimental results, in terms of the average cpWER metric, the LLM-based approach (Sys. 5, Tab. 5) not only outperforms the AED-based method using the same amount of data (Sys. 4, Tab. 5) but also remarkably surpasses the models in Sys. {1-3} that were trained with large-scale supervised data. It is worth mentioning that Sys. 1 in Table 5 was trained using 900k hours of supervised data, which is 1000 times more than what we used. This demonstrates that for SOT-based multi-talker ASR task, having a robust, large-scale pre-trained decoder is more important, as it provides strong capabilities in long-context awareness and cross-utterance modeling. This is precisely the advantage of LLM-based architectures over traditional AED-

Table 5. Overall performance comparison of various approaches on AMI-SDM evaluation set. Sys. {1-3} are previous works that use large-scale supervised data for pre-training. Sys. {4-6} display the results of models pre-trained with only 0.83k hours of LibriMix and then fine-tuned on AMI, where Sys. 4 uses the AED-based architecture, and Sys. {5-6} use the LLM-based architecture. The WER (%) and cpWER (%) metrics are reported for the *utterance groups* with different numbers of speakers, as well as overall (average) results.

Sys.	Architecture	Supervised Pre-training data	Fine-tuning data	WER (w.r.t. # of talkers) (%) ↓					cpWER (w.r.t. # of talkers) (%) ↓				
				avg.	1	2	3	4	avg.	1	2	3	4
1	Conformer AED [11]	900k hrs	AMI	-	-	-	-	-	21.2	14.7	19.6	**25.7**	**35.5**
2	Whisper medium [36]	680k hrs		-	-	-	-	-	23.6	12.8	21.8	32.5	45.9
3	Whisper large [36]			-	-	-	-	-	21.4	12.0	20.0	29.3	40.6
4	WavLM Large AED	**0.83k hrs**		30.5	16.7	26.2	45.8	54.8	24.1	10.8	20.4	37.9	48.6
5	WavLM Large LLM			27.6	14.9	25.3	38.4	52.6	21.0	9.3	18.8	31.1	44.1
6	+ beam search (beam=4)			**26.8**	**14.8**	**24.4**	**37.5**	**49.4**	**20.4**	**9.3**	**18.1**	30.3	42.2

Table 6. Speaker counting accuracy (%) for each *utterance group* of AMI-SDM evaluation set. The number of talkers can be estimated by counting the segments obtained by separating SOT-style transcriptions with the speaker change symbol "$". SOT follows speaker-wise FIFO, as shown in Fig. 1.

Sys.	Actual # of talkers	Estimated # of talkers					
		0	1	2	3	4	≥ 5
Tab. 5, Sys. 1	1	0.2	**97.2**	2.5	0.1	0.0	0.0
	2	0.0	13.7	**80.5**	5.9	0.0	0.0
	3	0.0	2.4	32.6	**60.2**	4.8	0.0
	4	0.0	0.0	9.9	51.2	**38.9**	0.0
Tab. 5, Sys. 4	1	0.0	**92.1**	7.6	0.3	0.0	0.0
	2	0.0	10.0	**73.0**	16.4	0.6	0.0
	3	0.0	0.8	30.2	**58.0**	10.1	0.9
	4	0.0	0.0	5.5	52.0	**33.5**	9.0
Tab. 5, Sys. 5	1	0.0	**96.7**	3.2	0.1	0.0	0.0
	2	0.0	11.7	**76.9**	11.0	0.4	0.0
	3	0.0	1.3	39.3	**48.4**	10.8	0.2
	4	0.0	0.0	10.5	53.0	**35.0**	1.5

based systems in a such complex scenarios involving multi-talker conversations. Additionally, the performance advancement of the LLM-based model over the AED-based method is further highlighted on the AMI evaluation set with an absolute WER reduction of 2.9% and cpWER reduction of 3.1% (Sys. 5 *vs.* 4, Tab. 5, column "avg.") comparing with the comparable systems evaluated on the LibriMix test set (Sys. 8 *vs.* 5, Tab. 1). This indicates that the more realistic and complex the scenario, the greater the advantage of the LLM-based method, confirming our conjecture. Using beam search for decoding yields even better results (Sys. 6, Tab. 5).

Comparing the results across *utterance groups* with different numbers of speakers, we find that the LLM-based method performs worse than Sys. 1 and Sys. 3 in groups with 3 and 4 speakers. This may be due to the limited supervised training data used in the LLM-based method, especially since the LibriMix dataset used for pre-training only contains two-speaker utterances, and the AMI training set has relatively few *utterance groups* with more than 2 speakers. In Sys. 2, the speech encoder in the Whisper medium model [26] has a parameter amount very close to that of WavLM Large, and the decoder of Whisper is also large. However, the LLM-based method consistently outperforms in *utterance groups* containing any number of speakers. This once again highlights the superiority of the LLM-based architecture, which leverages a powerful pre-trained decoder, over the AED-based architecture, where the decoder has not undergone specialized pre-training, in recognizing SOT-style long transcriptions with related content from multiple speakers.

We calculated the speaker counting accuracy and presented it in Table 6. From the results, it can be observed that the LLM-based method (Tab. 5, Sys. 5) is less accurate in estimating the number of speakers compared to the AED model trained with large-scale supervised data (Tab. 5, Sys. 1). Additionally, in the cases of 3 and 4 speakers, it also shows no significant advantage over the AED model using the same amount of data (Tab. 5, Sys. 4). Despite the lower accuracy in speaker counting, the LLM-based method achieves the best performance in the cpWER metric, indicating that it has a very high accuracy in recognizing the content of transcriptions in complex scenarios involving multi-talker conversations with noise and reverberation.

4. CONCLUSIONS

In this paper, we pioneer an LLM-based multi-talker ASR approach. In the evaluation, the proposed method achieves state-of-the-art results on both the simulated data LibriMix and the real-world data AMI, even outperforming existing methods trained with 1000 times more supervised data on the AMI-SDM evaluation set. The experimental results demonstrate that LLM-based architectures, which emphasize decoder performance and possess strong capabilities in understanding long contexts and modeling across utterances, outperform AED-based structures that focus more on encoder performance in SOT-based multi-talker ASR task. The LLM-based method has a much larger advantage on real data AMI than on simulated data LibriMix, which further highlights the potential of LLM-based models in handling speech processing tasks in complex and challenging scenarios.